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EE1060: Differential Equations and Transform Techniques

Lecture 30: Convolution and Kernels

Topics :

1. Kernels and Convolution
 2. Anti-causal kernels
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Kernel and Convolution - Notion of Inner Product

As a tool to compute system response:

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$y(t) = \int_{-\infty}^{\infty} \underbrace{h(\tau) x(t-\tau)}_{\text{inner product of } x(\tau) \text{ \& } h(t-\tau)} d\tau$$

convolution $\rightarrow$ also def for any signals $x(t)$ & $z(t)$

$\downarrow$

if $y(t)$ is chosen s.t. it transforms a given signal in a useful manner

$z(t)$ to be a kernel.

2 aspects about kernel

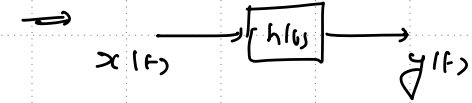
a) choice of kernel $[z(t)]$

b) Realization $\rightarrow$ come up with a system whose impulse response is $z(t)$

from Lec 29:- $h(t) = \underbrace{u(t)}_{\text{Realization (✓)}} \Rightarrow y(t) = \int_{-\infty}^t x(\tau) d\tau$

Causal kernel - Exponential Decay kernel

def: $\underline{h(t)} = \underline{g(t)} = e^{-at} u(t) \quad (a > 0).$



Realisable (?):- (✓) RL Ckt.

a) if $\underline{x(t) = u(t)}:-$

$$y(t) = \int u(\tau) \underbrace{h(t-\tau)}_{\text{causal}} d\tau.$$

$$= \int_0^t e^{-a(t-\tau)} d\tau.$$

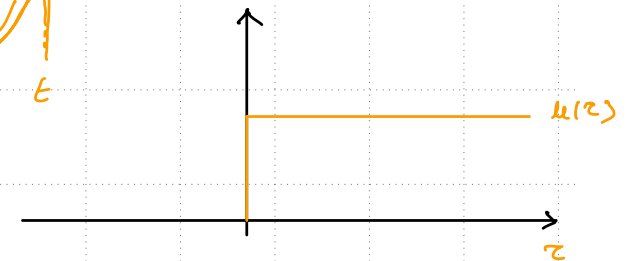
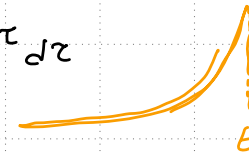
$$= e^{-at} \int_0^t e^{a\tau} d\tau$$

$$y(t) = \left(\frac{1 - e^{-at}}{a} \right) \text{ for } t > 0.$$

$$y(t) = 0 \text{ for } t < 0$$

$$\underline{h(t-\tau)} = \begin{cases} e^{-a(t-\tau)} & \tau \leq t \\ 0 & \tau > t \end{cases}$$

$$\underline{u(\tau)} = \begin{cases} 1 & \tau \geq 0 \\ 0 & \tau < 0 \end{cases}$$



b) if $\underline{x(t) = e^{-\alpha t} u(t)}:-$

$$\alpha \neq a$$

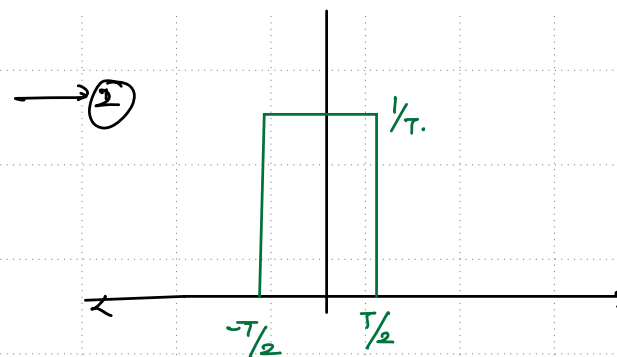
$$\alpha = a.$$

b) if $\underline{x(t) = \sin \omega_0 t u(t)}:-$

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Anti-Causal Kernel - Box Kernel

def:
$$h(t) = \begin{cases} 1/T & \text{for } |t| < T/2 \\ 0 & \text{otherwise} \end{cases}$$

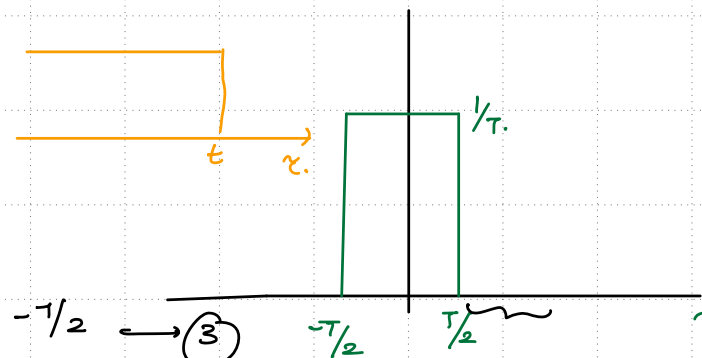


Realization? $\Rightarrow \times$ (anti-causal)

Unit step: $x(t) = u(t)$. $h(t) \rightarrow$ ②

$y(t) = h(t) * x(t)$

$$u(t-\tau) = \begin{cases} 1 & \tau \leq t \\ 0 & \tau > t \end{cases}$$



$$y(t) = \int_{-T/2}^t h(\tau) d\tau \quad \text{for } t \geq -T/2 \rightarrow$$

③

eval ③ for $t \leq T/2$

& for $t \geq T/2$ ($h(\tau) = 0$ for $\tau > T/2$ $\rightarrow$ upper limit $\approx T/2$)

Anti-Causal Kernel - Triangular Kernel

$$h(t) = \begin{cases} 1 - \frac{|t|}{T} & \text{for } |t| < T. \\ 0 & \text{otherwise} \end{cases}$$

Realization :- ??

